Hassaan Muhammad Khan

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linkedin.com/in/hassaan-muhammad-khan github.com/Hassaanmk hassaanmk.github.io/my-portfolio Scholar

ML Engineer bridging raw data and production-grade models. Passionate about Multimodal & NLP research, pushing the boundaries of how machines see and interpret human context.

Education

National University of Sciences and Technology (NUST) — *Karachi, Pakistan*
Bachelor of Electrical Engineering, AI Major — *Nov 2021 – Jun 2025*

- **CGPA: 3.89 / 4.0 Chancellor's Silver Medal** (2nd highest CGPA of the batch)
- **Thesis:** Autonomous SLAM-Based Interactive Robot
- **Coursework:** Artificial Intelligence, Machine Learning, Computer Vision, NLP, Data Structures & Algorithms, Networks, OOP, Digital Logic Design

Beaconhouse College *A-Levels | Physics A*, Mathematics A, Chemistry A* — *Aug 2019 – Oct 2021*
Beaconhouse School *O-Levels | 5A*, 4A* — *2014 – Jun 2019*

Experience

ByteCorp — *Cambridge, UK | Karachi, Pakistan*
Machine Learning Engineer — *Dec 2025 – Present*

- Implemented a multimodal pipeline integrating person recognition (YuNet, OpenCV) with Whisper-based audio transcription.
- Deployed stacked classifier for the Framingham Heart Study dataset achieving **83.5 AUC**.
- Integrated StrongSORT tracker with YOLOv12 for robust person tracking; deployed in Hyundai and MG showrooms, Saudi Arabia.
- Conducted research on Urdu discourse classification with the NLP research team.

Doaz — *Seoul, South Korea | Remote*
AI Engineer, NLP-Focused — *Oct 2025 – Dec 2025*

- Developed a multilingual AI Risk Checker (LangChain-based RAG system) on Korean chemical data from Doosan.
- Led AI team on Geotechnical AI Agent project from POSCO; delivered scaled VLM deployment achieving **92% inference accuracy**.

AI Intern — *Jun 2025 – Oct 2025*

- Conducted comparative analysis of RAG, CRAG, and GraphRAG, developing LangChain architectures from scratch.
- Deployed a FastAPI server to serve Hugging Face LLMs with custom ChatCompletion methods.
- Leveraged open-source VLLMs for scaled multi-lingual data extraction from PDFs.

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — *Abu Dhabi, UAE | Remote*
Visiting Researcher — *Aug 2024 – Jul 2025*

- Implemented Direct Preference Optimization (DPO) on a Llama fine-tuned model.
- Developed open-source **DRAG** framework for evidence-based retrieval via knowledge graphs, inspired by Microsoft's GraphRAG.
- Evaluated DRAG for knowledge distillation from LLMs to SLMs, improving accuracy on benchmarks by **10–27%**.
- Published at **ACL 2025 Main Conference**, Vienna, Austria.

Undergraduate Research Intern — *May 2024 – Jun 2024*

- Selected for UGRIP: highly competitive program with **4% global acceptance rate**; the only Pakistani intern.
- Conducted research under Prof. Zhiqiang Shen on prompt optimization for foundation models.
- Fine-tuned Llama 70B (sharded multi-GPU) via AWS, boosting accuracy by **∼10%** on MMLU, HellaSwag, and TruthfulQA.

Additech-Sim, German-based Startup — *Stuttgart, Germany | Remote*
Machine Learning Researcher — *Sep 2023 – Dec 2024*

- Integrated pre-trained model with custom RL environment (StableBaselines3), achieving **15% increase** in pressure prediction accuracy.
- Analyzed Mercedes-AMG experimental dataset on vehicle structural analysis.
- Enhanced parameter optimization using Genetic Algorithms and Variational Autoencoders.

Computer Scientist Intern — *Jul 2023 – Aug 2023*

- Deployed scikit-learn regression models to predict optimum pressure values; optimized data pipeline via EDA.

NUST AirWorks, Pakistan's First International Student UAV Team — *Karachi, Pakistan*
Automation Lead — *Aug 2022 – Jun 2024*

- Directed a 5-member department developing avionic systems and hardware interfaces for UAVs (hexcopters, fixed-wings, hybrid VTOL).
- Led end-to-end automation of X-VTOL, winning the **Performance Award at Teknofest 2023**, Turkey.

Senior Software Executive — *Dec 2021 – Jun 2024*

- Fine-tuned YOLOv8 for VTOL marker detection; deployed on Jetson Nano with GStreamer and OpenCV for real-time processing.
- Constructed path-planning algorithm and OpenCV AI for search-and-rescue using a swarm of decentralized drones.

Research Publications

DRAG: Distilling RAG for SLMs from LLMs to Transfer Knowledge and Mitigate Hallucination *ACL 2025 Main*
- Jennifer Chen, Aidar Myrzakhan, Yaxin Luo, **Hassaan Muhammad Khan**, Sondos Mahmoud Bsharat, Zhiqiang Shen
- Annual Meeting of the Association for Computational Linguistics, Vienna, Austria

Reinforcement Learning to Improve Finite Element Simulations for Shaft and Hub Connections *Stuttgart 2024*
- **Hassaan Muhammad**, Narmeen Sabah, Jan Falter, Markus Wagner, Boris Eisenbart, Matthias Kreimeyer, Muhammad Saeed
- Stuttgart Conference on Automotive Production 2024, ARENA2036, Stuttgart, Germany

Mitigating Data Scarcity in Low-Resource Discourse Classification via LLM-Driven Cross-Lingual Synthetic Generation *ACL ARR 2026, Under Review*
- Muhammad Owais Mahmood, **Hassaan Muhammad Khan**, Syed Ali Ahmed, Muhammad Ibrahim Chhipa, Bahram Khan Baloch

Selected Projects

DRAG Framework | *Knowledge Distillation, RAG, Graph-based Retrieval, SLMs*
- Knowledge distillation approach evaluated on MMLU, ARC-Challenge, Open-LLM leaderboard using SLMs. ACL 2025 Main.

Autonomous SLAM-Based Interactive Robot | *ROS2, SLAM, Jetson Nano, LLM Integration*
- Final Year Project: SLAM algorithms with ROS2 navigation stack and LLM interaction, deployed on single-board computers.

POSCO Drill-Log Application [private repo] | *Qwen-2.5-VL, fine-tuned YOLO, Tesseract OCR, Pydantic, Streamlit*
- AI-driven geological data extraction platform for drilling logs; fine-tuned YOLO with OCR achieving **92% accuracy**.

RAG Comparative Analysis | *LangChain, FAISS, Neo4j, GraphRAG*
- Built and compared RAG, CRAG, and GraphRAG architectures; deployed Neo4j knowledge graph for advanced query retrieval.

Doosan Safety Checker | *LangChain, Qdrant, Multi-lingual RAG*
- Multi-lingual RAG system on Korean regulations, accident records and chemical data using Qdrant vector store.

Facial Emotion Recognition | *CNN vs ViT, FER-2013, AffectNet*
- Comparative analysis of CNN and Vision Transformer architectures for facial emotion detection on FER-2013 and AffectNet.

RL for Finite Element Simulations [private repo] | *StableBaselines3, Custom RL Environment, Regression*
- Custom RL environment using supervised regression model for FEM shaft and hub pressure value inference.

CV-based RAG System | *PyMuPDF, Semantic Chunking, FAISS, Jina Reranking, Gemini / Groq, Streamlit*
- End-to-end RAG over CV PDFs with provider-swappable LLM, embedding and reranker backends configured via environment variables; live demo deployed on Streamlit Community Cloud.

Awards, Volunteering & Certifications

Awards
- **Chancellor's Silver Medal**, NUST: 2nd highest CGPA of the graduating batch *Jun 2025*
- **Rector's High Achiever's Award**, NUST *Aug 2022, Aug 2023*
- **Performance Award**, International UAV Competition, Teknofest, Turkey *2023*

Volunteering
- **IEEE Women in Engineering**, NUST Chapter: Senior Design Executive, lead organizer of inter-university events *Sep 2021 – Present*
- **The Citizens Foundation**: Volunteer, taught English and Theatre modules to underprivileged students *Jun 2023*

Certifications (Coursera / DeepLearning.AI)
- Supervised ML (Jul 2023) | Advanced Learning Algorithms (Dec 2024) | Improving Deep Neural Networks (Jul 2025)
- Structuring ML Projects (Sep 2025) | NLP: Classification & Vector Spaces (Sep 2025)
- NLP: Probabilistic Models (Jan 2026) | Gen AI: Beyond the Chatbot (Aug 2025)

Technical Skills

Languages	Python, C++, SQL
ML / DL	PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, ONNX, TensorRT
Computer Vision	YOLO (v8/v12), OpenCV, YuNet, StrongSORT, GStreamer, Jetson Nano, VLLM
NLP / LLM	LangChain, LangGraph, Transformers, NLTK, Cohere, Jina AI, LM-Evaluation-Harness, Fastchat
Voice AI	Whisper, Vapi, ElevenLabs
RAG / Vector	RAG, CRAG, GraphRAG, FAISS, Neo4j, Qdrant
Cloud / Deploy	AWS, Azure, Google Cloud, Vercel, FastAPI, Streamlit, Gradio, Vast.ai, RunPod, Postman
Tools	Git, Docker, npm, n8n, Pydantic, Outlines, Matplotlib, Seaborn, Plotly
Languages (Human)	Urdu (native); English: C2 Listening, C1 Reading, Writing, Speaking